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Nudibranch Diversity of South-East Pelion, Greece: An Eight-Year Citizen Science and Field Research Record (2017–2025)

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Abstract

We report 152 georeferenced observations of nudibranch molluscs (Gastropoda: Nudibranchia) recorded in the shallow coastal waters of South-East Pelion, Magnesia, Greece between June 2017 and June 2025. Systematic monitoring was conducted primarily via snorkelling and freediving at depths of 0.5–3.0 m across nine survey sites spanning approximately 9.5 km of coastline from Ammitsa to Platanias. A total of 13 confirmed species were identified, with 83% of records attaining Research Grade status on the iNaturalist citizen science platform under the BiodiversityGR project. The assemblage is dominated by aeolid nudibranchs (suborder Aeolidina, 93% of records), particularly *Cratena peregrina* (53 records, 40.2%) and *Flabellina affinis* (36 records, 27.3%). Observations are strongly seasonal, with 72% of records in June,

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consistent with Mediterranean phenological patterns. The Neraki sea cave complex near Katigiorgis village is identified as a biodiversity hotspot, yielding 9 confirmed species from 41 observations across four survey years, including a single confirmed record of the rare *Berghia verrucicornis* and evidence for a persistent population of *Nemesignis banyulensis*. The dataset represents the most extensive nudibranch occurrence record for the Pelion Peninsula and provides a baseline for ongoing marine conservation monitoring by Merman Conservation Expeditions Ltd.

Keywords: Nudibranchia; Aeolidina; Mediterranean; citizen science; Pelion;

1. Introduction

Nudibranchs (order Nudibranchia, class Gastropoda) are soft-bodied marine invertebrates widely distributed across the world's oceans from the intertidal zone to abyssal depths. In the Mediterranean Sea, the shallow rocky infralittoral supports a diverse nudibranch community closely linked to prey availability, particularly hydroid, bryozoan, sponge, and sea anemone colonies. Despite their ecological prominence as predators and bioindicators of benthic habitat quality, Mediterranean nudibranchs remain incompletely documented in many regions, including Greece, where systematic survey effort has been geographically fragmented.

The Pelion Peninsula in the Magnesia Regional Unit of Thessaly occupies a biologically significant position at the confluence of the Aegean Sea and the semi-enclosed Pagasetic Gulf. Its south-eastern coast features a mosaic of exposed rocky shores, sea caves, and sheltered coves, which support exceptionally clear water and diverse benthic communities. Despite these

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conditions, no prior nudibranch inventory for the area has been published.

This report documents nudibranch observations accumulated through a combination of opportunistic citizen science monitoring and structured field surveys between 2017 and 2025. The primary objectives are: (1) to establish a baseline species list for the nudibranch community of South-East Pelion; (2) to characterise spatial, temporal, and habitat patterns in the dataset; and (3) to identify priority sites for ongoing conservation monitoring.

2. Materials and Methods

2.1 Study area

The study area encompasses the south-eastern coast of the Pelion Peninsula, Greece (approximately 39°05′–39°15′N, 23°18′–23°25′E). Survey sites span from Ammitsa and Liri in the north to Platanias in the south, covering approximately 9.5 km of shoreline. The coastline is characterised by rocky subtidal substrates dominated by coralline algae, sponges, tunicates, and hydroids. Summer sea surface temperatures range from 22 to 28°C. Nine discrete survey sites were established: Neraki complex (including Neraki, Neraki Cave, and Neraki Beach), Radakia, Mikros Mourtias, Gianniotis, Agrielia, Ammitsa, Platanias, Vlachorema, and Liri Beach.

2.2 Survey methods

Observations were made during snorkelling and freediving sessions at depths principally between 0.5 and 3.0 m. All individuals were photographed in situ using waterproof digital cameras. No specimens were collected. GPS coordinates were recorded for each observation using a Nikon Coolpix W300 underwater camera. Observation depth was recorded for 98 of 132

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species-level records (74%) using an underwater camera. Sea surface temperature was not consistently measured.

2.3 Data recording and quality assessment

All observations were submitted to the BiodiversityGR citizen science project hosted on the iNaturalist platform (www.inaturalist.org) under observer account *ctaklis*. The iNaturalist quality pipeline assigns Research Grade status when two or more community identifiers agree on a species-level identification and no more than one-third of identifications disagree. Of 152 total records, 126 (83%) attained Research Grade status. The remaining 26 (17%) are classified as Needs ID, predominantly comprising nudibranch egg masses and individuals observed only at order or suborder level (*Nudibranchia, Doridina*), and are excluded from species-level analyses. Species nomenclature follows the World Register of Marine Species (WoRMS).

2.4 Survey period and observer effort

The dataset spans June 2017 to June 2025. Survey effort was highest in 2020 (62 observations) and 2021 (44 observations), corresponding to intensive local snorkelling activity during the COVID-19 pandemic period. From 2024 onwards, structured surveys under Merman Conservation Expeditions Ltd incorporated standardised search protocols within defined survey zones, though the dataset remains primarily opportunistic in character. The 2025 dataset comprises 17 records from a single survey session at the Neraki complex in June 2025.

3. Results

3.1 Species composition

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A total of 13 nudibranch species were confirmed across the survey period (Table 1). The assemblage is strongly dominated by aeolid nudibranchs (suborder Aeolidina), which account for 93% of species-level observations. Dorid nudibranchs (suborder Doridina) contribute 7% of records, represented by three species: *Felimare picta*, *Felimare villafranca*, and *Peltodoris atromaculata*. The most abundant species was *Cratena peregrina* (Wandering Cratena; 53 records, 40.2%), followed by *Flabellina affinis* (Mediterranean Violet Aeolid; 36 records, 27.3%), *Calmella cavolini* (12 records, 9.1%), and *Facelina rubrovittata* (8 records, 6.1%). The remaining nine species each contributed four or fewer records.

Table 1. Confirmed nudibranch species from South-East Pelion (2017–2025), ordered by observation frequency.

Species	Family	Suborder	Obs. (n)	Sites
Cratena peregrina	Facelinidae	Aeolidina	53	Neraki, Radakia, Agrielia, Mikros Mourtias, Ammitsa, Liri, Platanias
Flabellina affinis	Flabellinidae	Aeolidina	36	Neraki, Ammitsa, Mikros Mourtias, Liri, Vlachorema, Gianniotis
Calmella cavolini	Facelinidae	Aeolidina	12	Neraki Cave, Neraki, Gianniotis, Mikros Mourtias
Facelina rubrovittata	Facelinidae	Aeolidina	8	Neraki, Agrielia, Liri
Felimare picta	Chromodorididae	Doridina	4	Radakia, Agrielia
Caloria quatrefagesi	Facelinidae	Aeolidina	4	Neraki, Agrielia
Edmundsella pedata	Coryphellidae	Aeolidina	3	Neraki Cave, Neraki Beach
Felimare villafranca	Chromodorididae	Doridina	3	Neraki Cave
Nemesignis banyulensis	Facelinidae	Aeolidina	3	Neraki
Peltodoris atromaculata	Discodorididae	Doridina	2	Pounta, Vlachorema
Calmella gaditana	Facelinidae	Aeolidina	2	Neraki
Berghia verrucicornis	Aeolidiidae	Aeolidina	1	Neraki Cave
Spurilla neapolitana	Aeolidiidae	Aeolidina	1	Mikros Mourtias

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3.2 Temporal patterns

June accounted for 109 of 152 total observations (72%), reflecting both genuine biological seasonality and concentration of observer effort in early summer. Nudibranchs in the Mediterranean are most active during late spring and early summer when seawater temperatures exceed 20°C and prey hydroids reach peak colony density. Survey effort in July and August was markedly lower, consistent with the busiest period of the tourist season at this coastal location. The dataset spans observations from May through October, with minor records in July (19), August (16), September (3), and October (1). The peak survey years were 2020 (62 records) and 2021 (44 records), corresponding to intensive local monitoring during the COVID-19 pandemic. A summary of survey sites, including coordinates and habitat characteristics, is provided in Table 2.

3.3 Spatial distribution and survey sites

Figure 1 shows the distribution of all observations across the South-East Pelion coastline, with each confirmed species displayed in a distinct colour. Observations cluster along the rocky coastline from Katigiorgis northward to Liri, with an additional cluster at Platanias in the south. The Neraki complex (39°10′32.7″N, 23°20′45.7″E) emerges as the primary hotspot, with 41 observations representing 9 confirmed species from four survey years. Figure 2 presents individual species spatial distributions for the eight most frequently observed taxa. Notably, *Cratena peregrina* was recorded along the entire surveyed coastline, whereas *Felimare villafranca* and *Felimare picta* were restricted to southern sites with deeper rocky habitat.

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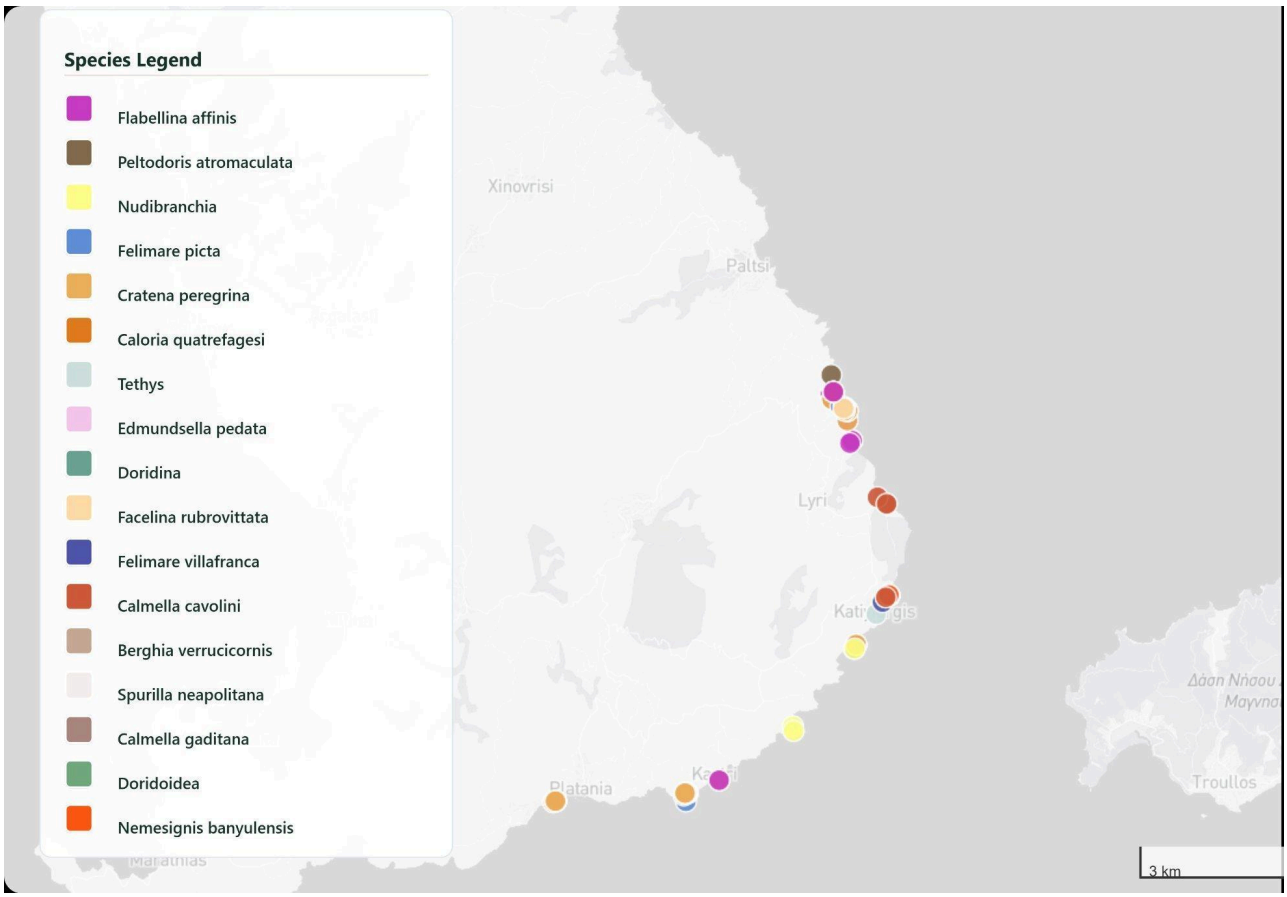

Figure 1. Distribution of all confirmed nudibranch species along the South-East Pelion coastline (2017–2025). Each colour represents a distinct species as shown in the legend.

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Figure 2. Spatial distribution of the eight most frequently observed nudibranch species across the South-East Pelion survey area. Observation counts shown in parentheses.

Table 2. Survey site summary: observations, confirmed species, and habitat characterisation.

Site	Obs.	Species	Coordinates (approx.)	Habitat notes
Neraki complex	41	9	39°10′33″N, 23°20′46″E	Sea cave, rocky overhang, 0.5–2 m
Radakia	16	4	39°08'24.6"N 23°17'55.1"E	Exposed rocky shore, 1–3 m
Mikros Mourtias	14	4	39°12'33.9"N 23°20'08.8"E	Semi-sheltered rocky cove
Gianniotis	14	2	39°12'31.3"N 23°20'09.4"E	Mixed rocky/sand, shallow
Agrielia	11	5	39°10'30.8"N 23°20'40.8"E	Mixed pebbles/rocky, 1–4 m
Ammitsa	11	4	39°12'44.5"N 23°19'58.8"E	Mixed pebbles/rocky, 1–3 m
Platanias	10	3	39°08'19.4"N 23°16'06.9"E	Sand/rocky shore, 1–2 m
Vlachorema	6	3	39°10'00.7"N 23°20'18.1"E	Sand/rocky cliff, shallow
Liri Beach	9	4	39°11'58.0"N 23°20'15.0"E	Mixed pebbles/sand

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3.4 Ecological observations and habitat associations

Of the 98 depth-recorded observations, 86% were from 1.5 m or shallower, with a modal depth of 1.0 m. This distribution reflects the snorkelling-based survey method but also indicates that the rocky infralittoral zone at 0.5–2.0 m supports the core nudibranch community. Aeolid nudibranchs were consistently recorded in direct association with hydroid colonies on vertical and overhanging rock surfaces. *Cratena peregrina* was primarily observed on colonies of *Eudendrium* spp., while *Flabellina affinis* was associated with *Coryne* and *Syncoryne* spp. Dorid species, including *Felimare picta* and *Peltodoris atromaculata*, were recorded from slightly deeper rocky faces (2.0–6.0 m) with encrusting sponge communities, consistent with their sponge-feeding trophic biology.

4. Discussion

4.1 Species richness and assemblage structure

The 13 confirmed species recorded here are broadly consistent with expectations for the shallow Mediterranean rocky infralittoral. The strong dominance of aeolid nudibranchs (93% of records) reflects both the high availability of hydroid prey in the survey area and the detectability bias inherent in snorkelling surveys, which favour shallow, active foragers over cryptic dorid species occupying deeper habitats. The presence of *Felimare picta* and *Peltodoris atromaculata* at depth indicates that a targeted SCUBA survey programme at 5–10 m would likely reveal additional dorid diversity, particularly at the sites Agrielia and Radakia, which support sponge-rich rocky walls at depth.

4.2 The Neraki complex as a biodiversity hotspot

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The Neraki sea cave complex stands out as a site of exceptional nudibranch diversity within the study area, yielding 9 confirmed species from 41 observations. The partially submerged cave at Neraki (GPS: 39°10′32.7″N, 23°20′45.7″E) supports a distinct microhabitat assemblage characterised by encrusting sponges, colonial tunicates, and thecate hydroids on dimly lit wall surfaces, conditions that support a community of cave-specialist taxa including *Calmella cavolini*, *Felimare villafranca*, and *Edmundsella pedata*. The single record of *Berghia verrucicornis,* an aeolid specialist feeding exclusively on sea anemones from the cave interior on 18 June 2020, is of particular biogeographic significance, as this species is rarely recorded in Greek waters and confirms sea anemone colonisation of the cave walls. The confirmed recurrence of *Nemesignis banyulensis* (Lady Godiva nudibranch) at Neraki in June 2021 and June 2025 strongly suggests a resident breeding population and warrants monitoring for population size estimation.

4.3 Seasonality and survey effort

The overwhelming concentration of records in June (72%) reflects a combination of genuine biological phenology and observer availability. Mediterranean nudibranchs are known to exhibit pronounced seasonal abundance peaks in late spring and early summer, driven by temperature-dependent hydroid prey cycles. The apparent decline in records from July onwards is consistent with literature reports of reduced nudibranch densities as summer temperatures peak and hydroid colonies senesce. However, the relatively few August–October observations in this dataset also reflect reduced survey effort during the tourist season, and the potential for late-summer species (e.g., *Caloria quatrefagesi*, recorded in September 2020 at Neraki) to be under-represented in the current dataset cannot be excluded.

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4.4 Citizen science and data quality

The BiodiversityGR project on the inaturalist platform proved an effective vehicle for accumulating georeferenced biodiversity records at relatively low cost. The 83% Research Grade success rate across 152 observations reflects both the photographic quality of field documentation and the willingness of the iNaturalist community to verify Mediterranean nudibranch identifications. The remaining 17% Needs ID records are concentrated in ambiguous aeolid pairs (e.g., *Flabellina* species-level discrimination) and egg mass observations, which are inherently difficult to identify to species without rearing to the adult stage. Standardised multi-angle photography protocols, as recommended in the conclusions below, would reduce this proportion in future surveys.

4.5 Conservation status and bioindicator value

None of the 13 confirmed species is currently listed as threatened under IUCN Red List criteria or protected under Greek national legislation. However, nudibranchs function as effective bioindicators of benthic habitat quality: their strict trophic specificity and sensitivity to water quality perturbations mean that assemblage changes can provide early warning of habitat degradation. The sustained presence of a diverse aeolid assemblage in South-East Pelion over eight survey years, including sensitive cave-specialist taxa, is a positive indicator of marine habitat quality in the region. The area falls within the broader Natura 2000 network context of the North Aegean islands complex, and its benthic biodiversity merits inclusion in regional marine conservation planning.

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5. Conclusions

This report documents the first systematic nudibranch inventory for the South-East Pelion Peninsula, establishing a baseline of 13 confirmed species from 152 georeferenced observations over eight years. Key findings are: (1) the assemblage is dominated by aeolid nudibranchs with strongest activity in June; (2) the Neraki cave complex is the primary diversity hotspot, yielding 9 species and rare records of conservation significance; (3) the species accumulation curve approaches saturation for the current survey method, though deeper surveys are expected to increase the species list; (4) the long-term dataset supports the use of this assemblage as a marine bioindicator for ongoing habitat quality monitoring.

Recommended actions for future monitoring include: formalising Neraki as a permanent monitoring site with photographic transects; increasing survey frequency to a minimum of two sessions per year (May/June and August); expanding depth coverage via SCUBA surveys at Agrielia and Radakia; engaging additional community observers via iNaturalist; and standardising photography protocols to improve species-level identification rates.

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Author Contributions

The author conducted all field observations, data collection, analysis, and manuscript preparation.

Ethical Compliance Statements

Conflict of Interest

The author declares that there are no conflicts of interest regarding the publication of this manuscript.

Ethics Statement (Field Observations)

This study is based exclusively on non-invasive field observations conducted in natural marine environments. No animals were captured, handled, or disturbed. Therefore, no specific ethical approval or permit was required under applicable Greek and EU legislation. All observations were conducted in accordance with standard ethical practices for wildlife research.

Data Availability Statement

The dataset supporting the findings of this study is publicly available in figshare:

Taklis, C. (2025). *An Open Data Contribution on Nudibranchs from the South-East Pelion Coastline*. https://doi.org/10.6084/m9.figshare.30217171

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